

Team 3 Capstone Project

Global Life Expectancy: Statistics → Dashboard → Cloud — R + Power BI + Azure

Team member	Primary focus on this project
Farah	Statistical analysis in R and write-up of findings
Filza	Power BI data model, DAX measures, and dashboard design
Hamna	Azure cloud hosting and publishing the report

Your toolkit: Statistical analysis in R, Power BI (DAX, data modeling, visualization), Azure / cloud

Your mission, in one sentence:

Find out, with real statistics, what actually drives how long people live — then turn that into a live, interactive dashboard anyone can open from a cloud link.

1. The problem you are solving

What actually drives how long people live in a country — money (GDP), schooling, healthcare spending, immunization coverage, or disease burden?

Your job is to answer that question rigorously using statistics, and then make your findings explorable by anyone through an interactive dashboard published to the cloud.

2. What you will build

A three-stage project: a real statistical study, an interactive dashboard, and a live published cloud report.

Stage 1 — R: multiple regression to find which factors significantly predict life expectancy, with assumptions properly checked and at least one hypothesis test (for example, comparing developed vs. developing countries).

Stage 2 — Power BI: an interactive dashboard built on a proper data model with DAX measures — filterable by country, year, and indicator.

Stage 3 — Azure: your data hosted in the cloud and your dashboard published to the Power BI Service with scheduled refresh, producing a live link anyone can open.

3. Your dataset

WHO Life Expectancy data covering about 193 countries from 2000 to 2015, with life expectancy plus roughly twenty health, economic, and social indicators (GDP, schooling, alcohol use, BMI, immunization rates, adult and infant mortality, HIV/AIDS, health expenditure, and developed-vs-developing status). This is a statistics dataset — your R analysis is the heart of the project, not just a data-loading step.

Download it here:

<https://www.kaggle.com/datasets/kumarajarshi/life-expectancy-who>

4. Step-by-step approach

1. In R, explore and clean the data, then run a multiple regression to see which factors significantly predict life expectancy.
2. Check your statistical assumptions (e.g. multicollinearity using VIF, residual normality) and run at least one hypothesis test, such as comparing developed vs. developing countries. Interpret your results honestly — don't just report that “everything correlates.”
3. Move your cleaned data to the cloud, for example using Azure SQL Database or Azure Blob Storage.
4. In Power BI, build a real data model and DAX measures, then design an interactive dashboard with filters, maps, and trend lines.
5. Publish your report to the Power BI Service and set up scheduled refresh, so it's reachable through a live cloud link.

5. Your timeline (2 weeks)

This capstone runs for 2 weeks. Pace yourself against the milestones below so you're not rushing at the end.

Week	Focus	You're on track if...
Week 1	Explore and clean the data in R; run the regression analysis, check assumptions, and run at least one hypothesis test.	Your statistical findings are complete, sound, and clearly interpreted in plain language.
Week 2	Move data to Azure and build the Power BI data model and dashboard. Publish to the Power BI Service, set up refresh, and finalize your write-up.	You have a live, shareable cloud link that works when someone else opens it, plus a completed write-up.

6. What to hand in

- An R analysis report with your regression results, assumption checks, and plain-language interpretation.

- The published Power BI dashboard — a live cloud link.
- A one-page note describing your cloud setup (where the data lives and how it refreshes).

7. How you will be graded

Your work will be assessed against the criteria below. Read this before you start — it tells you exactly what “good” looks like for this project, so there are no surprises later.

What we assess	What “strong” looks like
Statistical rigor (R)	Correct methods are used, assumptions are actually checked, and results are interpreted honestly.
Dashboard quality (Power BI)	A real data model with DAX measures and a clear, interactive design — not just charts pasted on a page.
Cloud deployment (Azure)	The report is genuinely published and reachable via a working cloud link, with refresh configured.
Storytelling	The dashboard answers the core question clearly, even for someone with no statistics background.

8. What a standout submission looks like

- A statistical finding that is genuinely insightful and well-defended, with assumptions properly checked and explained.
- A polished, intuitive dashboard that a policymaker could use without any training.
- A clean cloud setup where the published report refreshes automatically from the hosted data.

9. If you get stuck

If your regression results look strange, check for multicollinearity first — several of these indicators are naturally correlated with each other.

If Azure setup feels overwhelming, start with the simplest option (Blob Storage for the cleaned CSV) before attempting a full SQL Database.

Ask your program lead for a check-in at the end of Week 1 to sanity-check your statistical findings before you start building the dashboard.